Vaidehi Wagh

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Education

Carnegie Mellon University **Expected May 2026**
Master of Science in Robotics | GPA: 4.0/4.0 *Pittsburgh, Pennsylvania*

Coursework: Embodied AI, LLMs, Multimodal Learning, Computer Vision, Reinforcement Learning, Optimal Control, Medical Robotics
Teaching Assistant: Multimodal Machine Learning, Computer Vision

College of Engineering, Pune **May 2023**
B. Tech in Mechanical Engineering | GPA: 8.83/10 *Pune, India*

Research Experience

MetaMobility Lab October 2024 – Present
Graduate Research Assistant | Supervisor: Dr. Inseung Kang *Pittsburgh, PA*

- Deployed an **end-to-end human-in-the-loop personalization system** for a hip exoskeleton, deploying **VLMs & ASR** over a **distributed robotics architecture (Meta Aria glasses, GPU inference, Jetson control)** to convert **natural language feedback and egocentric vision** into structured controller updates. Incorporated **contextual bandits** for user profiling and **continual adaptation**.
- Engineered a **real-time multimodal inference system** with **2.67s end-to-end latency on exoskeleton** and developed a prior **on-edge voice-only control interface** that used **quantized LLMs (<3s latency, 8GB Orin Nano)** for offline intent parsing and adaptation.
- Built a **data-efficient temporal learning framework** (TCNs + transfer learning) achieving **+27% accuracy with <0.2% subject-specific data**, incorporating **cross-sensor synchronization (OpenSim + MATLAB APIs)** to align simulated and real IMU streams for robust temporal modelling in clinical populations.

Neuroplasticity, Imagery and Motor Behaviour Laboratory May 2022 – August 2025
Research Intern | Supervisor: Dr. Sarah Kraeutner *Kelowna, BC*

- Deployed a **motion tracking pipeline** using **OpenCV** and **MediaPipe** pose estimation to analyse upper-limb kinematics **(~0.14M actions, ~0.3M frames)**, enabling **high-throughput analysis** and identification of key clinical markers in stroke populations.
- Led a validation study comparing low-cost hand tracking to traditional motion capture using shape similarity analysis **(16K fine upper-limb motion trajectories, ~70% accuracy)** demonstrating a **scalable & resource-efficient solution** for dexterous movement analysis.

Publications

[1] **Wagh et al.**, "Transfer learning for biological joint moment estimation in stroke populations", American Society of Biomechanics 2025.

[2] **Wagh et al.**, "Using MediaPipe to track upper-limb reaching movements after stroke: a proof-of-principle study", Journal of NeuroEngineering and Rehabilitation, 2025. *(link)*

[3] **Wagh et al.**, "Quantifying Similarities Between MediaPipe and a Known Standard to Address Issues in Tracking 2D Upper Limb Trajectories: Proof of Concept Study", Journal of Medical Internet Research: Formative Research, 2024. *(link)*

Professional Experience

Deloitte Touche Tohmatsu Ltd – *Analyst, Enterprise Technology & Performance* August 2023 – June 2024

- Deployed cloud-based pipelines and visualization tools for **large-scale financial time-series data**, implementing temporal forecasting and analytics to automate reporting and generate actionable insights.

LightRay Technologies (*Early-stage startup*) - *Computer Vision Intern* November 2022 – May 2023

- Fine-tuned a **Tiny-YOLOv4** model in PyTorch on a custom 12k image dataset, achieving **94.6%** class-wise accuracy, improving **real-world object detection** in complex environments and enabling **on-device deployment**.

Projects

LLM based human-robot voice interface – *Talking to Robots* August 2025 – December 2025

- **Finetuned Qwen3-VL-2B (SFT + LoRA)** on a custom-generated dataset to enable **long-horizon planning** using **multimodal vision and audio** inputs for wearable robot control.

Automated IV insertion Bot – *Medical Robotics* August 2025 – December 2025

- Designed and 3D-printed **custom linear actuators**, **developed & integrated robot arm control** and **vision-based vein detection** pipelines, using **real-time robotic system integration** to reduce procedure time by **23%.**

Generative Adversarial Imitation Learning – *Introduction to Robot Learning* January 2025 – April 2025

- Used **inverse RL** with **state-level Bidirectional GANs** and **trajectory-level GAIL** for learning humanoid locomotion rewards from PPO generated expert walking trajectories.

Skills

Programming & ML: **Python (PyTorch, OpenCV, HuggingFace, Llama-cpp, Ollama), Quantization, E2E Deployment, Real-time Inference,** MATLAB, R, Julia, Git, Docker, Linux, WSL | **Robotics/Embedded:** NVIDIA Jetson, Arduino, ESP32, Raspberry Pi | **Design & Simulation:** SolidWorks, Fusion 360, Inventor, AutoCAD, OpenSim, MuJoCo